

Name: Aashish **Course:** BCA in Cyber Security

Task: Create a Strong Password and Evaluate Its Strength

1. Objective

Understand the factors that contribute to a strong password and test them against industry-standard password strength tools.

2. Password Strength Testing

Using the tool *passwordmeter.com*, the following evaluations were performed:

Password	Complexity Factors	Result/Score
Ashish123	Simple (lowercase + numbers)	Weak
As#2026_	Mixed case, numbers, symbols	Medium
mohan@2026_	Length, complexity, and symbols	Strong
ShriRAdhey@delhi1005355	Length, complexity, Numbers and symbols	Very Strong

WEAK PASSWORD

PasswordMonster

info@passwordmonster.com

How Secure is Your Password?

Take the Password Test

Tip: When adding a capital or digit to your password, don't simply put the capital at the start and the digit at the end

Show password: ☒

Ashish123

Very Weak

9 characters containing:

Lower case

Upper case

Numbers

Symbols

Time to crack your password:
63.02 seconds

Review: Oh dear, using that password is like leaving your front door wide open. Your password is very weak because it contains 2 dictionary words and a sequence of characters.

Your passwords are never stored. Even if they were, we have no idea who you are!

MEDIUM PASSWORD

PasswordMonster

info@passwordmonster.com

How Secure is Your Password?

Take the Password Test

Tip: When adding a capital or digit to your password, don't simply put the capital at the start and the digit at the end

Show password: ☒

As#2026_

Medium

8 characters containing:

Lower case

Upper case

Numbers

Symbols

Time to crack your password:

6 days

Review: Hmm, using that password is like locking your front door, but leaving the key under the mat.

Your password is of medium strength because it contains a dictionary word and a combination of characters that are close together on the keyboard.

Your passwords are never stored. Even if they were, we have no idea who you are!

STRONG PASSWORD

PasswordMonster

info@passwordmonster.com

How Secure is Your Password?

Take the Password Test

Tip: When adding a capital or digit to your password, don't simply put the capital at the start and the digit at the end

Show password: ☒

mohan@2026_

Strong

11 characters containing:

Lower case

Upper case

Numbers

Symbols

Time to crack your password:

22 days

Review: Good, using that password is like locking your front door and keeping the key in a safety deposit box.

Your passwords are never stored. Even if they were, we have no idea who you are!

VERY STRONG PASSWORD

PasswordMonster

info@passwordmonster.com

How Secure is Your Password?

Take the Password Test

Tip: When adding a capital or digit to your password, don't simply put the capital at the start and the digit at the end

Show password: ☒

ShriRAdhey@delhi1005355

Very Strong

23 characters containing:

Lower case

Upper case

Numbers

Symbols

Time to crack your password:

16 trillion years

Review: Fantastic, using that password makes you as secure as Fort Knox.

Your passwords are never stored. Even if they were, we have no idea who you are!

3. Key Concepts & Definitions

- **Brute Force Attack:** An attack where the perpetrator uses software to guess every possible combination of characters until the password is found.
- **Dictionary Attack:** An attack that uses a list of common words or previously leaked passwords to gain unauthorized access.
- **Password Complexity:** Increasing security by using a mix of uppercase, lowercase, numbers, and symbols.
- **Password Length:** A critical factor; longer passwords are significantly harder to crack.
- **Multi-Factor Authentication (MFA):** A security layer requiring two or more verification methods.

- **Passphrases:** Long, sentence-like passwords that are easier to remember but highly secure.

4. Best Practices Learned

- **Never use personal information (names, birthdays) in passwords.**
- **Always maintain unique passwords for every account to prevent cascading breaches.**
- **Utilize a password manager to handle complex, randomized credentials.**
- **Regularly update passwords and enable MFA on all sensitive accounts.**

1 What makes a password strong?

A strong password consists of a minimum of 12-16 characters, combining uppercase letters, lowercase letters, numbers, and special symbols. It avoids dictionary words or personal information to prevent easy guessing.

2 What are common password attacks?

The most common methods include Brute Force, where attackers systematically guess combinations, and Dictionary Attacks, which use a pre-compiled list of likely passwords.

3 Why is password length important?

Length is the most critical factor in password security because it exponentially increases the number of possible combinations a hacker must check, significantly slowing down cracking attempts.

4 What is a dictionary attack?

A dictionary attack is a targeted approach where software attempts to access an account by cycling through a list of common words, phrases, and previously compromised passwords.

5 What is multi-factor authentication (MFA)?

MFA is a security process that requires users to provide two or more verification factors to gain access to an account, such as a password plus a temporary code sent to a mobile device.

6 How do password managers help?

Password managers help by generating and storing unique, complex passwords for every individual account, which eliminates the need to reuse passwords and prevents the risk of one breach affecting all accounts.

7 What are passphrases?

Passphrases are long strings of words or sentences that are easier for humans to remember but much harder for computers to crack due to their extreme length and complexity.

8 What are common mistakes in password creation?

Common mistakes include using easily guessable information like birthdays or names , reusing the same password across multiple platforms, and choosing passwords that are too short to resist automated attacks.